

IoT Introduction

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UGE

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Outline

1 Introduction

2 Networks and Protocols

- Wireless Short Range
- Wireless Long Range
- wired protocols
- Devices

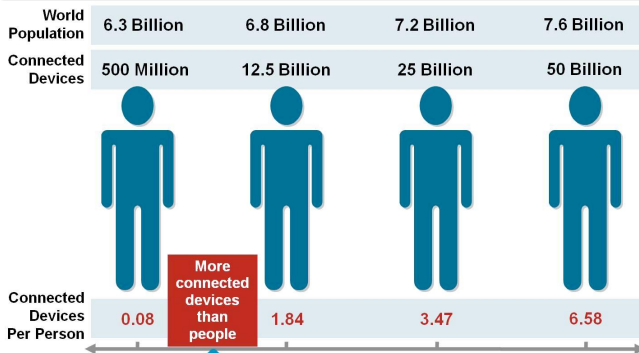
3 Approachs and Architectures

4 From Objects to Applications

What is IoT ?

some remarks

- 1 Buzz word
- 2 Not very successful
- 3 More devices connected than humans living (year 2011)
- 4 50 billions devices connected in 2020



6/83



A definition

Internet of Data

- 1 from Server to Computer
- 2 Content for human reading (text image video)
- 3 (except web services)

Internet of Things

- 1 Internet : public protocols for interconnection
- 2 Things : devices with connectivity and processing capabilities

A comparison

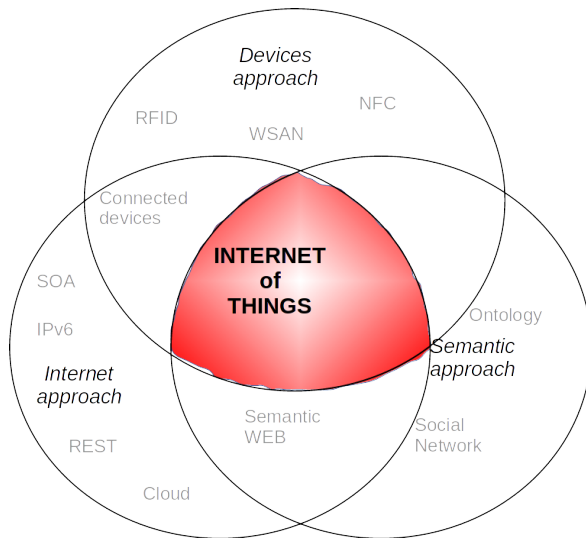
Internet of Data

- 1 from Server to Computer for human reading
- 2 Less servers that users
- 3 high heterogeneity of document types
- 4 few documents, big size

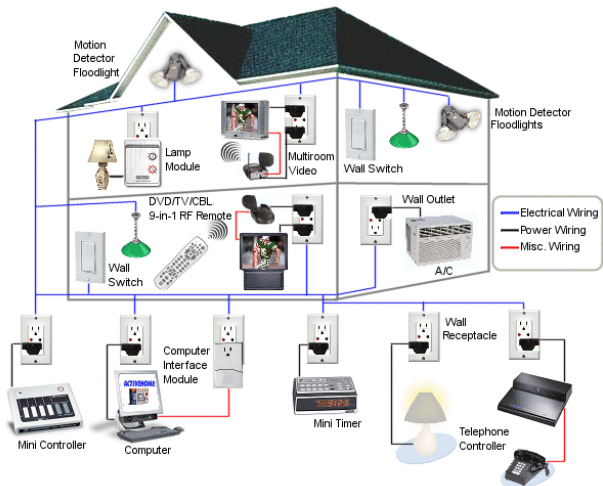
Internet of Things

- 1 Things to Cloud (or Things to Things)
- 2 very small data, not versatile
- 3 send often

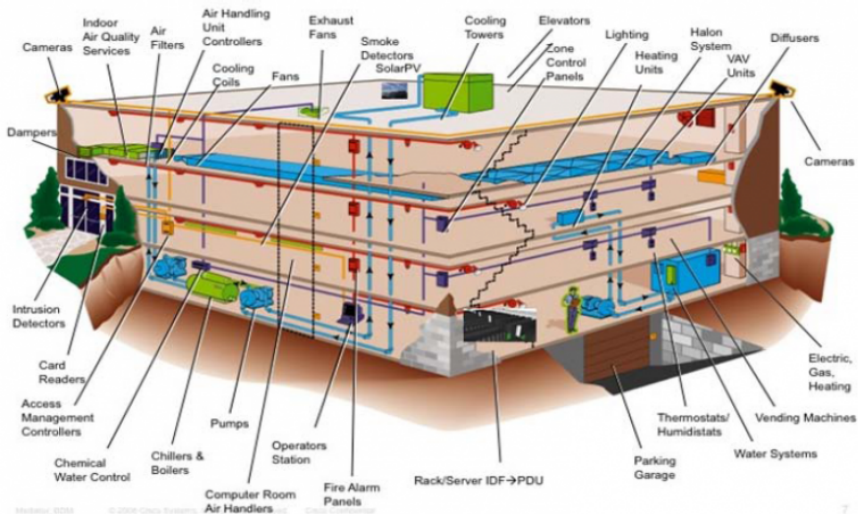
So many point of view



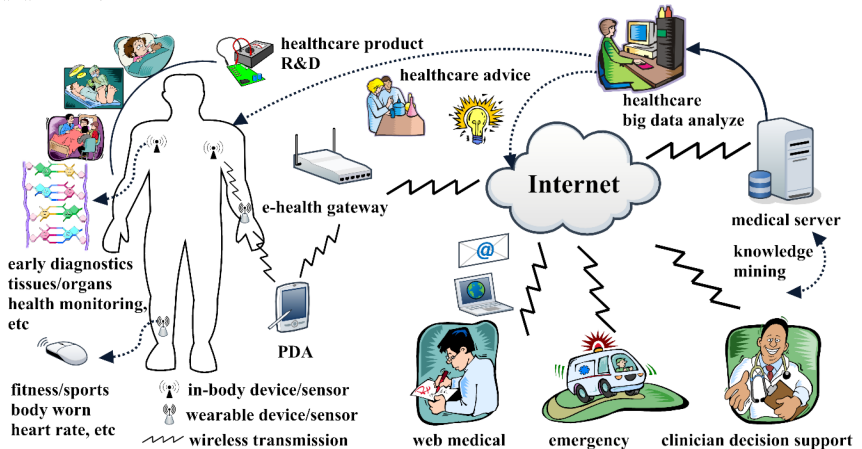
Many usage: SmartHome



Many usage: Smart Building

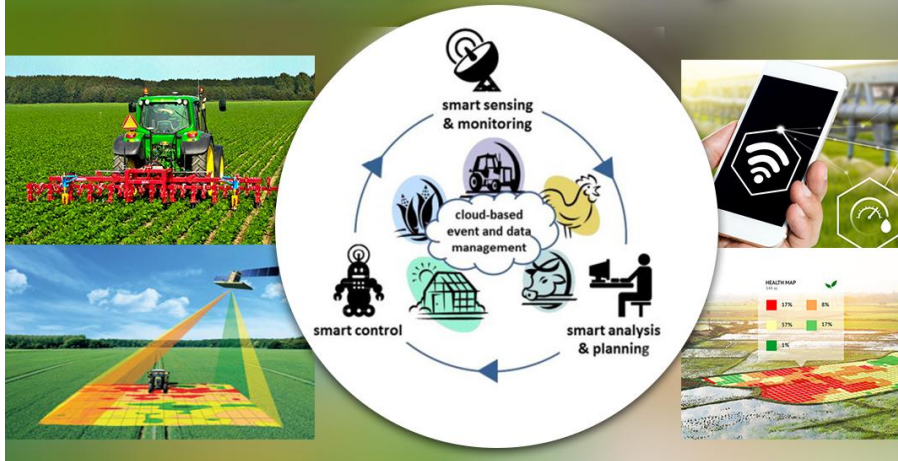


Many usage: e-Health



Many usage: Smart Farming

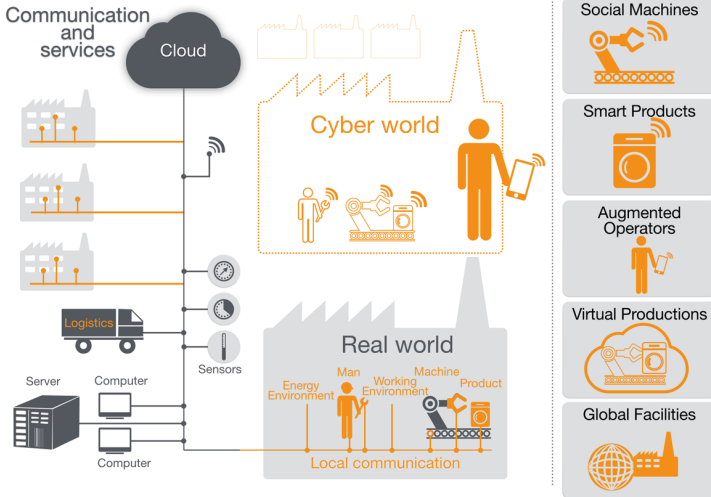
IoT in Smart Farming



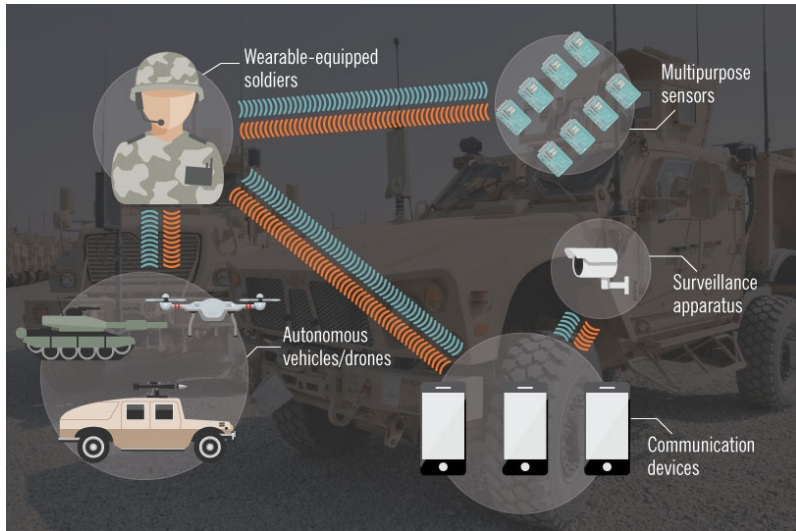
Many usage: Smart Transportation



Many usage: industry 4.0



Many usage: Smart Military



Tools, Systems, Network

Welcome in Babel

- 1 Physical : Ethernet, FO, ADSL, 3G/4G/5G, WAN, PAN, LowPAN, Bluetooth, RFID, NFC
- 2 User Equipment OS : MS, Apple, Android, Contiki-OS, Linux, TinyOS, RIOT, FreeRTOS, Mantis, LITEOS, thinksquare, Brillo...
- 3 Programming Languages : C, NesC, Java, Threads ou pas...
- 4 Data Format : JSON, XML, SenseXML, EXI....
- 5 Protocols : Low layer : 6LoWPan, ZigBee, 4G, 5G, Sigfox, LoRa - Higher layer : CoAP, AMQP, MQTT, XMPP, DPWS, WEAVE

A more complete definition

Internet of Things

- 1 Interaction between digital and real world
- 2 Capillary Internet
- 3 Extension of the Internet to Things and Places of the physical world

Elements of the IoT

IoT Objects

- 1 able to measure (sensor) or modify (actuator) the world
- 2 able to communicate
- 3 able to process data

IoT Passive Objects

- ① Passive, but identifiable

Infrastructure element

- 1 Gateway
- 2 Servers

Elements of the IoT

IoT Applications

- 1 Fuzzy definition
- 2 Often seen as "remote control"
- 3 Often very centralized (big data)
- 4 No common Architecture/Framework

What about ?

- 1 M2M, D2D, V2V, V2I ?
- 2 Industry 4.0 ?
- 3 Autonomous ? Pervasive ?
- 4 Distributed computing (edge, cloudlet, mist) ?

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡

Perspectives : Standard de facto ?



"Silicon.fr " (2017/04/17)

Perspectives : Another fighter

Perspectives : And the Winner is not...

A screenshot of the Google Cloud IoT Core announcement page. The page features the Google Cloud logo at the top left, a search icon, a language selector, and a user profile icon. A prominent yellow banner contains the announcement text in French. Below the banner, the heading 'IoT Core' is displayed in a large, bold font. Underneath the heading, a paragraph describes the service as a solution for connecting, managing, and securely ingesting data from IoT devices. A blue button labeled 'Trouver un partenaire' is positioned below the text. At the bottom, a link to the documentation is provided. The overall design is clean and professional, typical of a corporate announcement page.

Examples: some personal objects

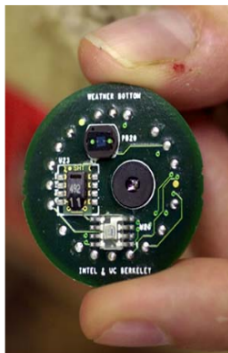


- 1 Philips Hue (lamp)
- 2 AWOK (lamp speaker)
- 3 Smart Watch
- 4 Nano computers
- 5 Smart devices

conferring digital
identity to things

Example: Environment

In the UC Berkeley Botanical Garden, 50 “micromotes” sensors are dangled like earrings from the branches of 3 redwood trees to monitor their growth.



A network of Objects for forest surveillance

Example: Smart City

The
Mercury
News

San Francisco's trash bins get smart

To avoid overflow, sensor-based system to let city know when cans are full



To improve urban waste management

Example: Smart City

Denver Police to Test New Gunshot Detection System

Police plan to conduct the tests this week from 6 p.m. to midnight by firing a series of gunshots into a bullet trap, according to Denver police. Officials say there will be no danger to the public.

BY SAJA HINDI, THE DENVER POST / AUGUST 26, 2019

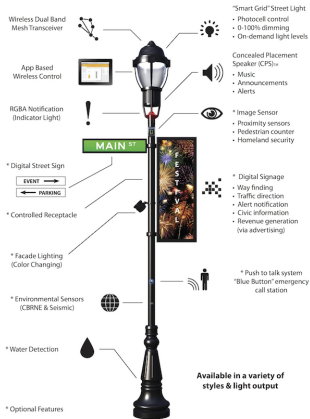


Using data analysis and event gathering for security purpose

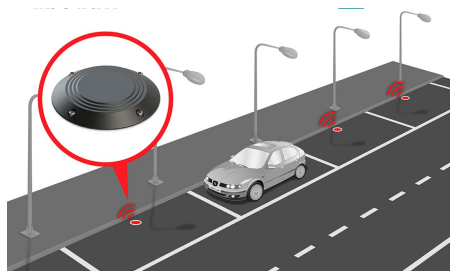
Example: Smart City

intelli**streets**SM

The wireless digital infrastructure that controls energy management, security, and outdoor media



Available in a variety of styles & light output



IoT Introduction (lesson 1)

1 Introduction

2 Networks and Protocols

- Wireless Short Range
- Wireless Long Range
- wired protocols
- Devices

3 Approachs and Architectures

4 From Objects to Applications

Wired and Wireless Internet

- 1 Fiber
- 2 xDSL
- 3 Ethernet
- 4 Wifi

Wired Objects

- 1 KNX

Wireless Objects

- 1 Short Range : EnOcean, ZWave, 802.15.4 (Zigbee, 6LoWPAN), Bluetooth Low Energy
- 2 Long Range : Lora, Sigfox, NB-IoT, LTE-M, 5G

Different Topologies

Star Topology



Tree Topology



Mesh Topology

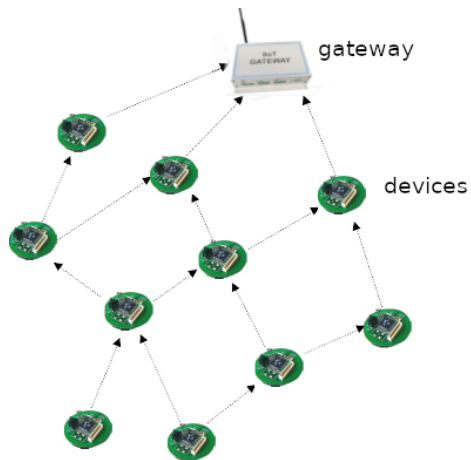


● WSN End Node ● WSN Router ● WSN Gateway

Wireless Objects

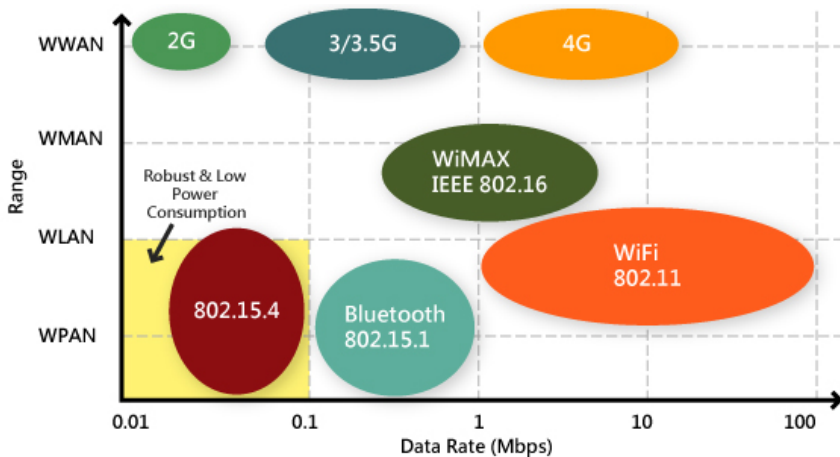
- 1 Star: every object connected to the center
- 2 Tree: hops to the sink through nodes acting as router
- 3 Mesh: every router may use several paths

Wireless Sensor Networks



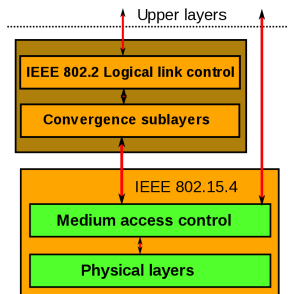
Self organized network of devices: no infrastructure.

Wireless technologies comparison



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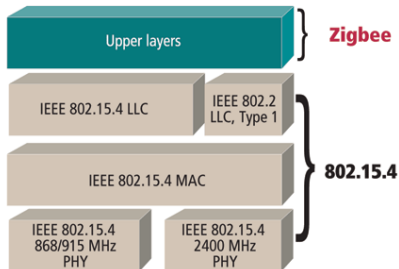
IEEE 802.15.4



802.15.4 Low Energy Consumption

- 1 Wireless Short Range (10 meters)
- 2 127 bytes frame max
- 3 250 kbits/s

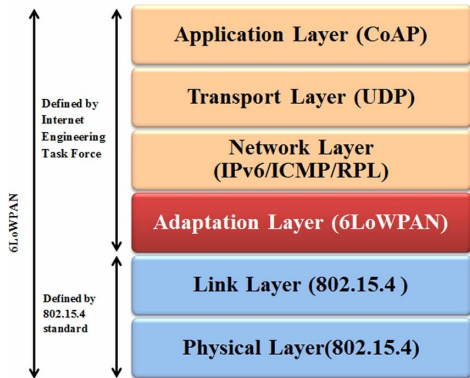
IEEE 802.15.4 and ZigBee



ZigBee

- 1 Wireless Short Range
- 2 Specific upper layer
- 3 Devices Profile

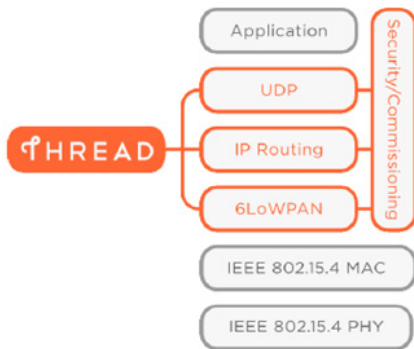
IEEE 802.15.4 and IETF 6LoWPAN



6LoWPAN

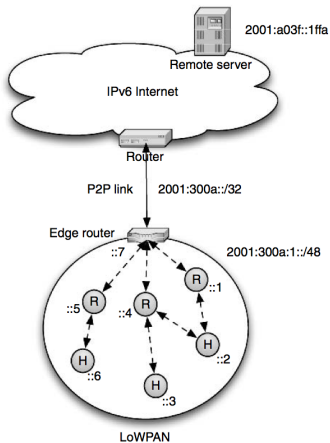
- 1 IEEE RFC
- 2 IPv6 portage over 802.15.4
- 3 End-to-end UDP/IP

IEEE 802.15.4 and Thread



Thread

- 1 Industrial Usage of 6LoWPAN
- 2 Google, Apple, Next, Samsung, ARM, Qualcomm...



Wireless Short Range

BlueTooth IEEE 802.15.1



BlueTooth

- ❶ 10 meters, 1Mb/s
- ❷ 2.4 Ghz radio band
- ❸ 1 master, up to 7 slaves

BlueTooth Low Energy (BLE)



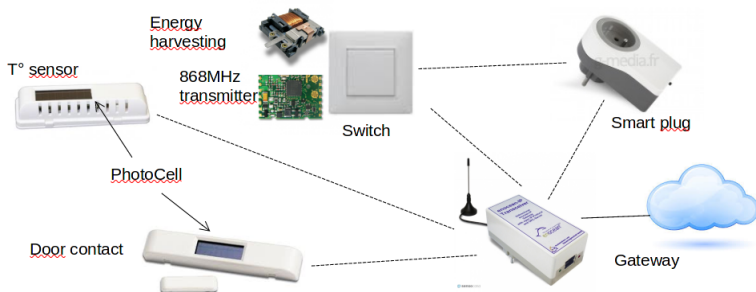
BlueTooth 4.0

- 1 60 meters, still 1Mb/s
- 2 Low energy consumption
- 3 1 master, up to 7 slaves



- 1 Central Controller
- 2 50 meters max, 9 - 40 Kbits/s
- 3 Security

EnOcean



EnOCean

- 1 Energy harvesting
- 2 lack of security

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Long range: the Cellular approach

Constraints LPWAN (Low Power Wide Area Network)

- 1 Energy is the issue
- 2 is short range a solution ? Need for a gateway!
- 3 Operators' Cellular networks cover the globe
- 4 Price ? Energy ? Bandwidth ? Not adapted to IoT objects and usages (nb of objects, not operated by human, regular, and not so sporadic data consumption, pushing data more than consuming data)

Constraints

- 1 Answer : Specific networks, or specific subsets of existing standards (2G, 4G)

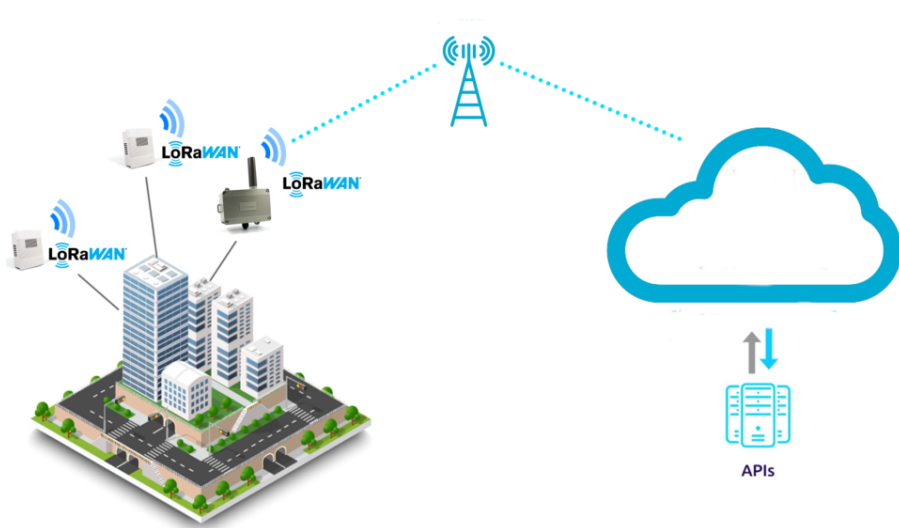
Specific solutions : LoRa and Sigfox

pionners in long range iot network

- ① main idea: Reuse 2G network for low power objects
- ② very low bandwidth, very long range
- ③ Create a private network, that can be then connected to the Internet (access through an API)
- ④ throughput : Lora 50kb/s, Sigfox 0.1kb/s
- ⑤ Sigfox : 140 msg a day (0 to 12 bytes), max range 50 kms - receive up to 4 msg a day (8 bytes)
- ⑥ Lora : 500 msg a day (up to 51 bytes), max range 15 kms - depending a class (among 3) more or less received msg (vs energy cost)

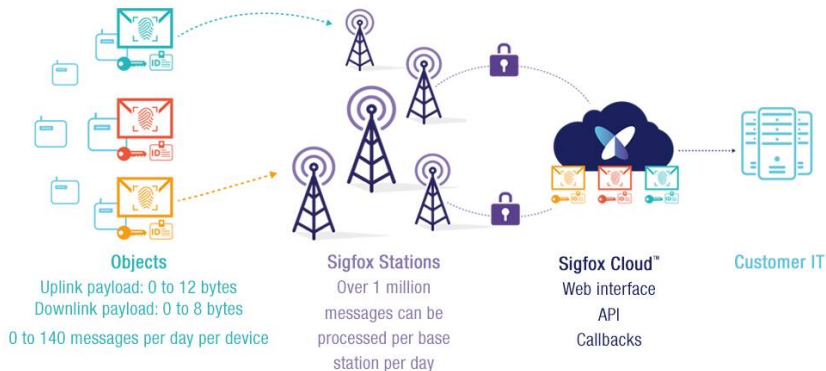
Wireless Long Range

LoRa



Wireless Long Range

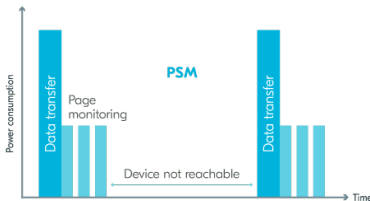
Sigfox (the end of the story ? (2022))



4G: Trying to catch the flag

taking over the Market

- 1 4G is wellspread over multiple countries
- 2 Idea: Reserve a part of the bandwidth for IoT
- 3 Propose new MAC to limit energy consumption (PSM Power Saving Mode (no access, full stop) and eDRX extended Discontinuous Reception (40x more sleeping period))
- 4 compared to real 4G, less bandwidth but longer range



Narrow Band Iot

NB-IoT



- ① based on 4G
- ② Bandwidth 200kHz (instead of 20 Mhz)
- ③ 60 (up to 169)kb/s upload, and 30 (up to 127)kb/s download
- ④ no mobility, adapted to static low power low throughput devices

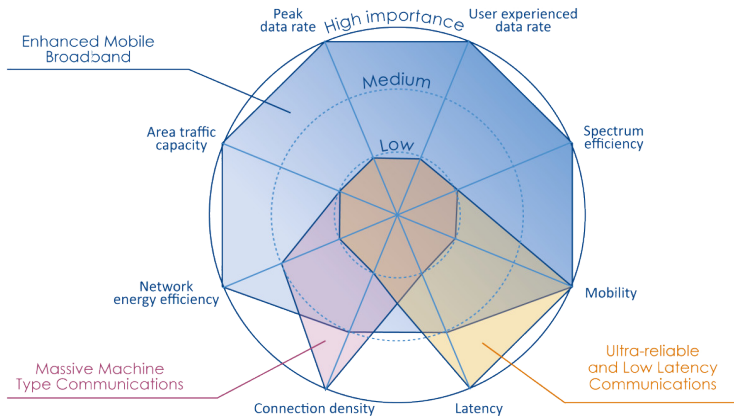
LTE-M

TCP/IP for 4G IoT



- ① based on 4G
- ② Bandwidth 1,4MHz (instead of 20 MHz)
- ③ 375kb/s upload, and 300kb/s download
- ④ (100kb/s for real with IP) Preferred mode
- ⑤ TCP + TLS available
- ⑥ Mobility, low latency, even voice and video capabilities

5G: a new hope ?



Enhanced Mobile Broadband



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 - **wired protocols**
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Wire: A solution ?

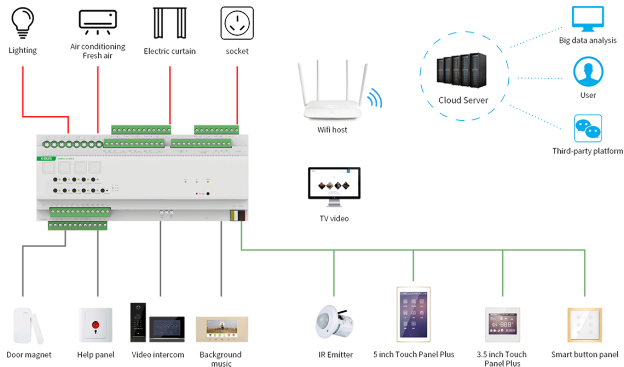
Issue with wireless

- 1 Energy !!
- 2 Latency, Bandwidth
- 3 Reliability, Interference

Wire

- 1 May not be available
- 2 Stability
- 3 Provide Communication + Energy
- 4 Speed, Bandwidth
- 5 Powerful objects

KNX



KNX for IoT

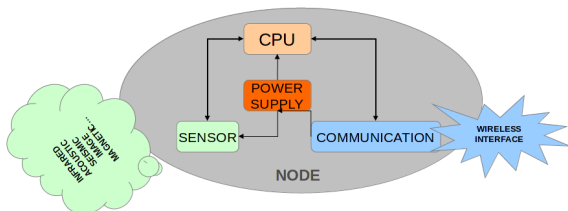
- 1 PowerLine + data bus
- 2 9600bps / Twisted pair
- 3 16 backbones of 16 zones of 64 modules

Power over Ethernet PoE

PoE standards

IEEE standard	PoE Type	Power to port	Maximum current
802.3af	Type 1	15.4W	350mA
802.3at	Type 2	30W	600mA
802.3bt	Type 3	60W	600mA
802.3bt	Type 4	100W	960mA

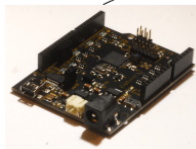
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Modules

- 1 Sensing/Actuating: interaction with real world
- 2 CPU: Processing, packet building, network mgmt
- 3 Transmission: Radio or wired communication
- 4 Energy control: idle, sleep or active mode

WSAN Nodes: 802.15.4



Microcontroller	Atmega128RFA1
Operating Voltage	3.3V
Input Voltage (recommended)	7-18V
Input Voltage (maximum)	6-30V (transients to -20V and +60V)
Digital I/O Pins	14 + 3 auxiliary
PWM Output Pins	6
Analog Input Pins	6 (0-1.8V)
I/O Protection	±30V transient
	-2.5V to +5.8V continuous
DC Current per I/O Pin	20 mA
DC Current for 5V Pin	250 mA
DC Current for 3.3V Pin	200 mA ²
Flash Memory	128 KB of which 2 KB is used by the bootloader
SRAM	16 KB
EEPROM	4 KB
Clock Speed	16 MHz
RF transmit power	+3.5 dBm
Receiver sensitivity	-100 dB
Antenna gain	2 dBi
Current Draw	30 mA (transmitting, USB, no I/O connections)
	15 mA (transmitting, no USB, no I/O connections)
	6 mA (radio off, no USB, no I/O connections)
	250 µA (sleep) ⁴



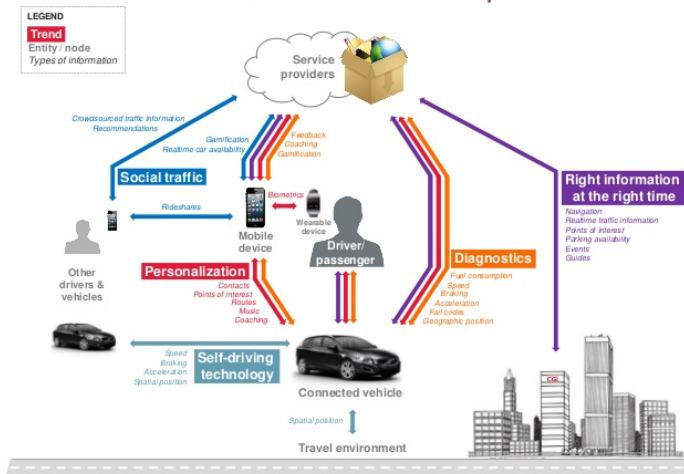
Various IoT devices, connected to the network, sending data, and receiving orders to execute.

100

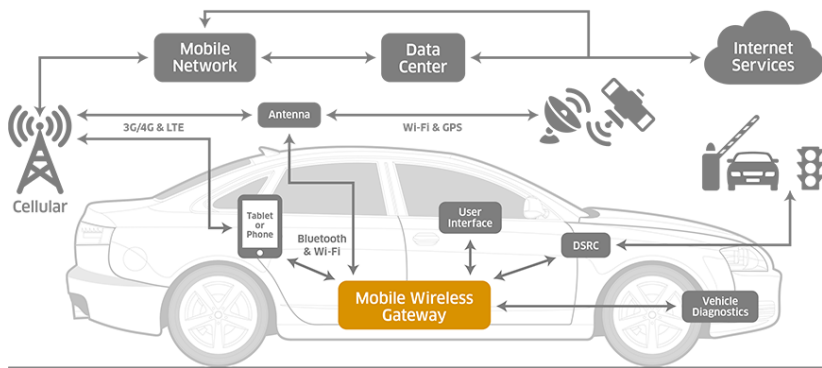


Vehicles

Smart vehicle trend map



Vehicles - 2



IoT Introduction (lesson 1)

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State of the art

Low Layers work

- 1 Many devices
- 2 Many networks
- 3 Many data
- 4 Many different areas covered (from body to country)

Main Issue: How to aggregate everything ?

- 1 Tools ?
- 2 Architecture ?
- 3 Interconnecting IT ?
- 4 Usages ?

Characterisation

Characterising your needs

- 1 your usage ?
- 2 your environment ?

Examples

- 1 area size
- 2 number and kind of devices
- 3 fixed or/and mobile, dynamic items
- 4 public/private, referenced object or discovered
- 5 data-centric, event-centric, user-centric
- 6 ...

Silos

Main trend: Silos

- 1 vertical solution, from devices to results
- 2 often closed
- 3 API ?
- 4 reusable, shareable ?

Main Issues

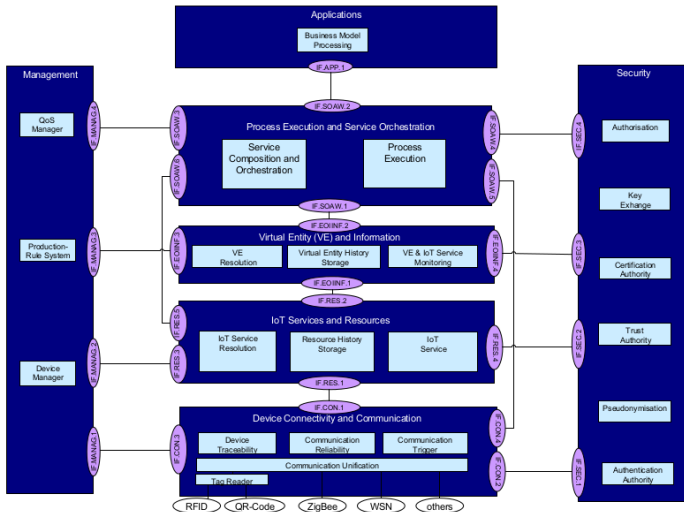
- 1 Big business, big money
- 2 Poor agility
- 3 customer in jail

IoT-A : a european proposition

IoT-A: a proposed architecture

- 1 An example of solution to interconnect silos
- 2 A Demo: [IoT-A video](#)
- 3 <http://www.iot-a.eu/public>

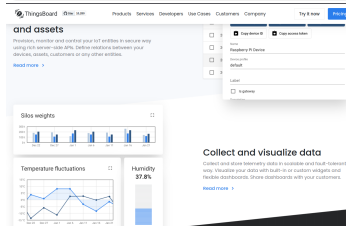
Inside the architecture



The first approach

Gather and represent data

- 1 Need to centralize and give access to a unique dashboard
- 2 Free software : Thingsboard.io <https://thingsboard.io/>



Itron: An example of a working solution

A metering infrastructure

- 1 Sensors for water, Electricity, gas...
- 2 Collect, qualify and store data
- 3 Dashboards

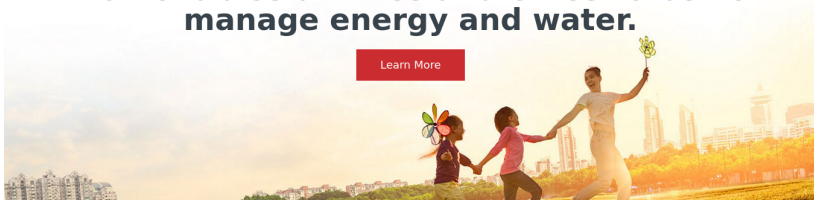
EMEA Blog Careers Investors Sign In Contact Q



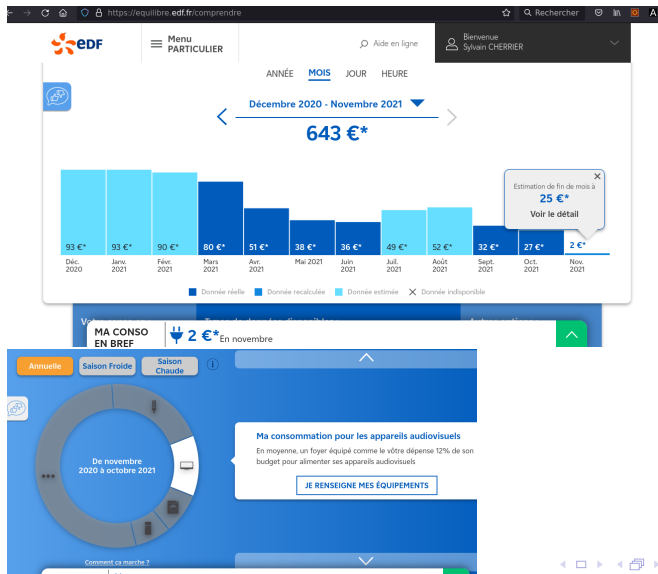
Why Itron Solutions Customers Partners Developers Resources Company

Itron enables utilities and cities to better manage energy and water.

Learn More



EDF : Another monitoring example



State of the art

As in the previous chapter :

Layers

- 1 Many different devices and networks
- 2 Many use-case, many data
- 3 For what purpose ? With which tools ?

Main Issue: How to aggregate everything ?

- 1 Depends on User point of view
- 2 A Specific application ? Supervision ? Remote Control ? Autonomous ?

Collecting Data

- 1 First need : gathering data
- 2 Send them somewhere: Which periodicity ?
- 3 How to handle service failure ?
- 4 Continuous or sporadic access ?
- 5 need for remote access ?

Qualifying Data Payload

- ① So many different objects
- ② measures: precision, range, unity
- ③ **Profile** helps to qualify data (and decode payload)

Date	Source	Valeur	Fcnt	Rssi, Snr, Esp, Sf	Nb Gateway	Tags	Signal réseau
02/11/2021 15:39:38	devEUI : 70B3D580A015DC6C8 deviceld : um:lo:nsid:lora:70B3D580A015DC6C8	09dc823a8a015dc30100	4010	-112 / -14.75 / -126.89 / 12	1	-	
02/11/2021 15:28:41	devEUI : 70B3D580A015DC6C8 deviceld : um:lo:nsid:lora:70B3D580A015DC6C8	09dc82018a015dc30000	4009	-113 / -9.75 / -123.19 / 12	1	-	
02/11/2021 15:13:06	devEUI : 70B3D580A015DC6C8 deviceld : um:lo:nsid:lora:70B3D580A015DC6C8	09dc840a8a015dc430000	4007	-114 / -4 / -119.46 / 12	1	-	

Measure data

<u>ID</u>	<u>Description</u>	<u>Type</u>	<u>Unit</u>	<u>Range</u>
<i>temperature</i>	Measured temperature	FLOAT	°C	-47...+128
<i>humidity</i>	Measured relative humidity	INT8	%	0...100
<i>luminosity</i>	Measured ambient luminosity	INT8	lux	0..60K
<i>presence</i>	Presence detection status during last period	UINT8	Bool	0..1
<i>battery_current_level</i>	Battery level of the device	UINT8	%	1...100

Payload	SenLab H Port 3 Datalog
----------------	--------------------------------

Description	Datalog Port 3 Temperature Humidity
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15
16	16
17	17
18	18
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93	93
94	94
95	95
96	96
97	97
98	98
99	99
100	100

Example :

Input 0384193C017C2A022B0A2C062D032E

Output

```
{
  "battery": {
    "level": {
      "unit": "%",
      "value": 51
    },
    "temperature": {
      "unit": "°C",
      "currentTemperatures": 25,
      {
        "timestamp": "2017-11-07T17:14:00.000Z",
        "value": 23.1875
      },
      {
        "timestamp": "2017-11-07T17:15:06:00.000Z",
        "value": 23.6875
      },
      {
        "timestamp": "2017-11-07T17:17:00.000Z",
        "value": 23.75
      }
    },
    "humidity": 11-07T17:13:00.000Z",
    "value": 46.0
  },
  {
    "timestamp": "2017-11-07T17:14:00.000Z",
    "value": 45.0
  },
  {
    "timestamp": "2017-11-07T17:16:00.000Z",
    "value": 43.0
  },
  {
    "timestamp": "2017-11-07T17:17:00.000Z",
    "value": 43.0
  }
}
```

Output ison fields :

How to deal with Data : storing, analyzing

The needs

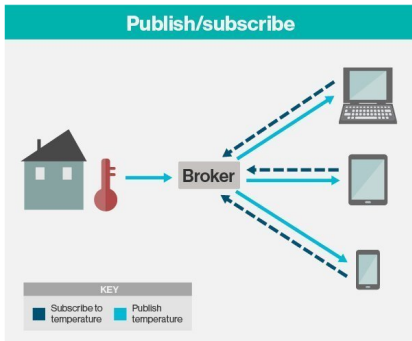
- 1 2 points of view: immediate reaction or post analysis
- 2 in any case, big storage...
- 3 ... of very small data

Solutions

- 1 Specific storing tools
- 2 With query language
- 3 connected to analysis tools

Storing : MQTT, AMQP

- 1 publish subscribe (leverage the objects network)
- 2 lightweight (for constrained network : bandwidth)
- 3 TCP/IP based
- 4 Normalized (OASIS and ISO)
- 5 3 level of services (QoS)



Cloud : Storing and analyzing

- 1 Lot of solution with Cloud Services
- 2 Still the amount of data is huge
- 3 Property ? Ownership ?
- 4 new trend: Edge storing, Edge computing



Solutions

News



Serving the customer : Applications.. and then ?

- 1 Mainly Dashboard
- 2 But is the future in Automation ?? (Industry 4.0 ?)
- 3 Simple example : IFTTT

What can IFTTT do?

To put it simply, anything!

